

Amritanshu Kumar

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Professional Summary

Backend engineer specializing in real-time distributed systems, with 3.5 years of professional experience. Owns and operates a live multiplayer platform serving 41,000 registered and approximately 3,400 daily active users across Android and iOS. Took the platform from a single instance to horizontally scaled: Redis-backed publish/subscribe fan-out, cross-instance leader election, distributed locking, and failover with zero session loss. Maintains 3,332 automated tests under enforced coverage gates. Leads a 7-engineer team spanning client, server, and API.

Technical Skills

Languages: TypeScript, JavaScript, C#, C++, Go, Python

Backend: Node.js, Express, Socket.IO, WebSockets, REST APIs, MongoDB, Redis, ioredis, JWT authentication, Winston

Distributed Systems: Horizontal scaling, Redis publish/subscribe, leader election, distributed locking, failover and draining, state synchronization, idempotency and deduplication, split-brain resolution, server-authoritative protocols

Cloud and DevOps: Google Cloud Platform, Google App Engine, Docker, GitHub Actions, CI/CD, structured logging, live operations, maintenance windows

Testing: Mocha, Chai, Sinon, c8 code coverage, end-to-end test harnesses, test-driven development

Other: Unity, Unreal Engine, System Architecture, Real-Time Multiplayer, Technical Leadership, Code Review, Mentoring

Professional Experience

Team Lead, Backend Architecture

Dec 2024 – Present

RootHoot Pvt Ltd

Remote, Hyderabad, India

- Own the Node.js and TypeScript socket server and Express REST API for Pongz end to end, a live real-time multiplayer platform running on Google Cloud across Android and iOS.
- Scaled the server horizontally from a single instance using a Redis publish/subscribe adapter for cross-instance event fan-out, per-instance identity, and room-routed delivery, so any node can reach a socket held by any other node.
- Solved matchmaking split-brain across N instances with Redis leader election. The elected leader forms matches from a shared global queue rather than each node's local slice, with atomic per-player claims that make double-seating impossible under overlapping leases.
- Built multi-instance failover with zero session loss. A draining node hands live games to its peers, which adopt them mid-session instead of dropping players. Validated by a two-instance end-to-end test suite running against real Redis.
- Engineered a server-authoritative protocol that eliminated race conditions caused by latency and message reordering, using monotonic sequencing, fingerprint-based deduplication on reconnect, and exponential backoff with jitter.
- Built live operations infrastructure including graceful maintenance windows with coordinated shutdown, player draining, and admission gating, so deployments land without terminating in-flight sessions. Also built the competitive ranking and skill-based matchmaking systems.
- Grew the automated test suite to 3,332 passing tests with CI-enforced coverage floors of 80 percent lines, 85 percent functions, and 70 percent branches, including deterministic real-socket and multi-instance end-to-end harnesses.
- Directed architecture, code review, and sprint planning across client, server, and API.

Software Engineer

Feb 2023 – Oct 2024

DeftSoft

Mohali, Punjab, India

- Shipped a product to more than 5 million downloads and a 4.6 star rating on Google Play. Iterated on production telemetry after launch to lift engagement by 30 percent.
- Introduced test-driven development across the team to catch regressions before release, which improved ratings by 1.5 stars. Authored a 25-chapter technical course on software architecture.
- Profiled and optimized hot paths for 25 percent higher throughput on low-end hardware, and built procedural tooling that cut content production time by 40 percent.
- Mentored 3 junior engineers through code review, pair programming, and coding standards.

Selected Projects

Nimirta Engine – 2D physics engine written from scratch in C++ and SFML. Impulse solver with warm starting, Baumgarte stabilization for high-velocity contacts, semi-implicit Euler integration, fixed timestep with spiral-of-death prevention, 3 collision algorithms, hand-written vector math. github.com/nimritagames/NimirtaEngine

InstaReload – Hot-reload toolchain that cuts reload latency from 3-5 seconds to 7 milliseconds. Runtime IL patching via Roslyn for the common path covering 90 percent of edits, with MonoMod detours for structural changes. github.com/nimritagames/Insta_Reload

FlowUI – Open-source UI framework under the MIT license that eliminates UI boilerplate. CRTP builders, Roslyn source generation with partial classes, bidirectional $O(1)$ lookups, and an 8-tab visual editor. github.com/nimritagames/Unity-FlowUI

Education

Bachelor of Computer Application

Sep 2020 – Jun 2023

Lovely Professional University, Punjab, India

Certifications: Unity Certified Programmer; Unreal Engine Game Design and Development Specialization